

OpenJustice Benchmarkathon Operational Guide

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1. What Is the OpenJustice Benchmarkathon?

The OpenJustice Benchmarkathon is a five-day, fully remote collaboration designed to test a new no-code method for integrating legal reasoning into AI systems.

Traditional AI benchmarks are typically built by technical teams and measure performance on narrow metrics such as multiple-choice answer accuracy. These evaluation frameworks are often made to assess the capabilities of AI in isolation, not how it would help lawyers in the real world.

This initiative takes a different approach. Rather than evaluating AI in isolation, we are examining how legal professionals use structured reasoning tools in practice and what makes this process effective, practical, and sustainable.

2. Shaping How Legal AI Is Built

AI systems do not determine what legal reasoning looks like. Lawyers do.

The quality, sustainability, and legitimacy of legal AI depend on whether legal professionals are actively involved in shaping how knowledge is encoded, structured, and evaluated.

The Benchmarkathon is therefore not only a technical exercise. It is a collaborative effort to:

- test a method for integrating structured legal reasoning into AI
- define evaluation standards grounded in real practice
- explore how legal knowledge can function as a shared professional resource
- assess how practitioners contribute to and interact with knowledge-driven AI systems

Beyond technical performance, we are examining whether this model of contribution is intuitive, efficient, and sustainable for legal professionals.

The central question guiding this initiative is straightforward:

Does explicitly encoding practitioner reasoning improve the reliability and adaptability of AI systems used in law?

3. Objectives (Build + Benchmark)

The Benchmarkathon combines three objectives:

1. **Building legal reasoning**, contributed by practicing lawyers.
2. **Benchmarking AI performance**, by testing how leading AI systems perform when guided by that reasoning.

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3. **Proposing end products** with engineers that shape how these reasoning flows would be used in real legal practice.

The event is not about making toy examples. It is a structured empirical test of whether encoding legal judgment improves the reliability and adaptability of AI systems in real legal practice.

4. What Is OpenJustice?

OpenJustice is a community infrastructure developed at the Conflict Analytics Lab (Queen's University) in collaboration with academic and judicial partners. Its purpose is to help lawyers encode structured reasoning and experiential knowledge into AI systems beyond simply adding more unstructured text.

Most AI tools rely on large volumes of documents and probabilistic pattern recognition. Legal and dispute resolution practice, however, depends heavily on structured reasoning:

- Balancing key factors (e.g., assessing reasonable notice based on age, role, tenure, and market conditions)
- Applying legal tests and thresholds (e.g., determining just cause or material breach)
- Sequencing procedural steps (e.g., when to negotiate, escalate, or litigate)
- Recognizing negotiation patterns and risk signals
- Translating legal analysis into client-facing advice

OpenJustice provides a pipeline for translating that structured and tacit knowledge into formats that AI systems can meaningfully use.

In practical terms, this means turning reasoning into explicit, testable structures — making legal knowledge more complete, more transparent, and more usable by AI systems.

The Benchmarkathon is designed to evaluate whether this method improves reliability and adaptability across different AI models.

5. Why This Matters: Reliability and Adaptability

Legal AI has captured widespread attention. Yet a fundamental question remains:

How can AI systems reliably function within the realities of legal practice?

Most existing systems are trained on large volumes of text and evaluated on case-law style questions. However, everyday legal work involves far more than retrieving doctrine.

Three gaps remain significant.

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a. Reliability in High-Stakes Contexts

Legal work often requires long-horizon, multi-step reasoning, balancing competing factors, applying structured legal tests, and interpreting incomplete or ambiguous facts.

Error rates remain significant in these contexts. In high-stakes matters - litigation strategy, risk assessment, transactional advice - probabilistic outputs are not sufficient.

Improving reliability requires more than better prompting. It requires encoding how legal reasoning actually unfolds.

b. Adaptability to Real Practice

A large portion of lawyering depends on tacit and experiential knowledge:

- how negotiations typically evolve
- how risk is calibrated beyond doctrine
- how procedural choices shape outcomes
- how advice is adapted to clients and counterparties

This dimension of practice is difficult to capture in traditional AI benchmarks. If AI is to support legal professionals meaningfully, it must adapt not only to legal texts, but to legal judgment.

6. What Participants Will Do

Participants will contribute their expertise to the Benchmarkathon in two structured components:

1. **Build Reasoning Flows**
2. **Test and Evaluate Reasoning Flows Through Simulations**
3. **(Optional) Integrate Reasoning Flows into Practical Tools**

These contributions will allow us to evaluate how AI systems perform when guided by explicit legal reasoning.

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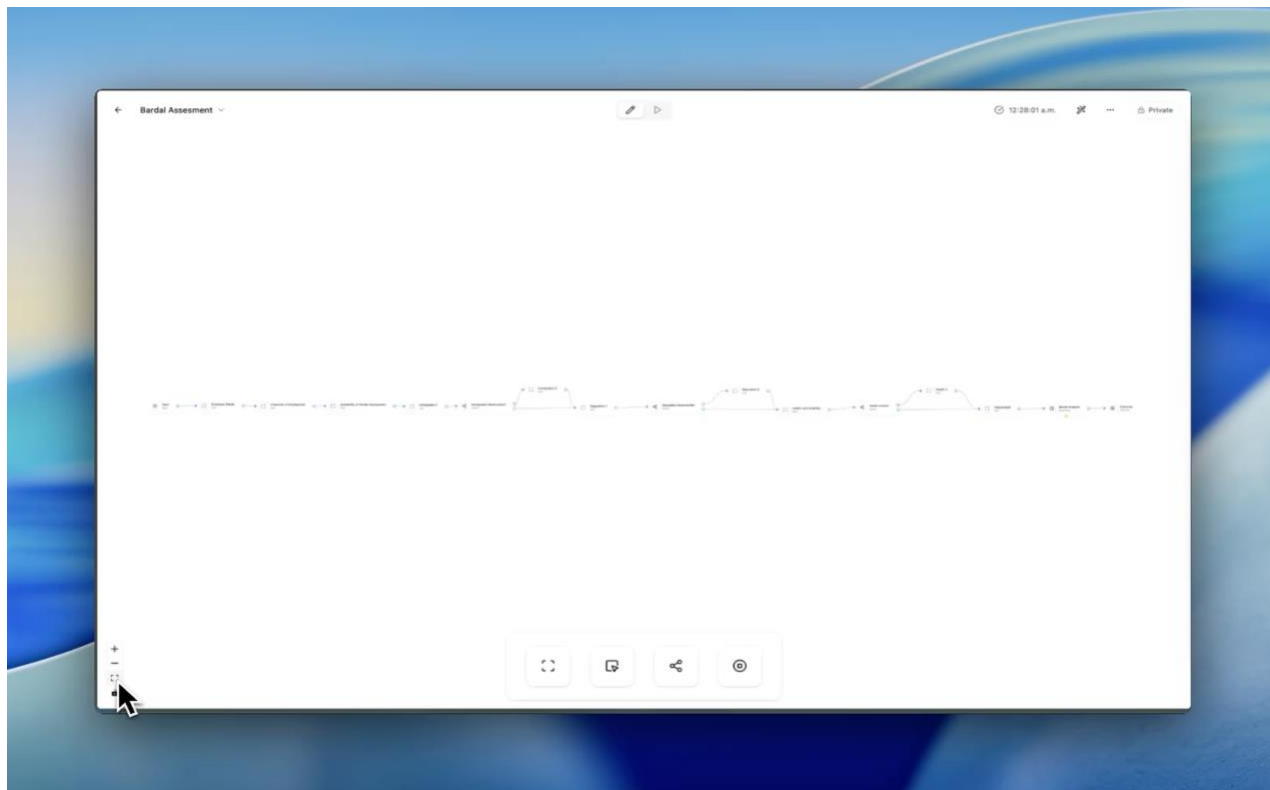


Figure 1: Bardal Reasoning Flow Demo

1. Reasoning Flows

A Reasoning Flow captures how you think through a recurring legal issue in your practice.

Participants are invited to select one issue they regularly encounter and describe:

- The key factors you consider;
- The legal tests or thresholds that apply;
- How those factors influence the outcome; and
- When and why the conclusion might change.

The goal is to make explicit the structured reasoning that normally remains implicit in professional judgment.

Examples include (see *OpenJustice User Guide*):

- Assessing reasonable notice in termination
- Evaluating litigation risk

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- Deciding when to escalate or settle a dispute
- Structuring a negotiation strategy in a recurring matter
- Calibrating settlement ranges based on legal and relational factors
- Designing mediation sequencing and offer framing

2. Test Cases

Each Reasoning Flow should be accompanied by at least five realistic case scenarios.

Think of these as practice-based examples that allow us to test how AI systems apply your reasoning whether in litigation, transactional work, negotiation, or dispute resolution.

Each test case should include:

The input: A fact pattern, timeline, or raw materials (e.g., emails, contracts, correspondence, mediation briefs, settlement offers, screenshots). This should represent your realistic starting point, or whatever materials you think would test AI systems the most.

The expected output: The **ideal professional deliverable** you would produce, such as:

- a legal memo or risk assessment
- client advice summary
- litigation strategy note
- settlement range analysis
- negotiation strategy outline
- mediation plan or caucus sequencing framework
- contract drafting, markup, or review

These test cases will allow us to compare how different AI systems perform when guided by your structured reasoning.

3. End Product

We invite engineers to come and build with our OpenJustice API, and how it integrates with existing AI tools. Some examples:

- Use OpenJustice to curate a comprehensive list of jurisprudence for a legal document drafting tool
- Create a strategic negotiation chatbot based on a corpus of settlement data
- Redline contracts with supporting regulations and caselaw

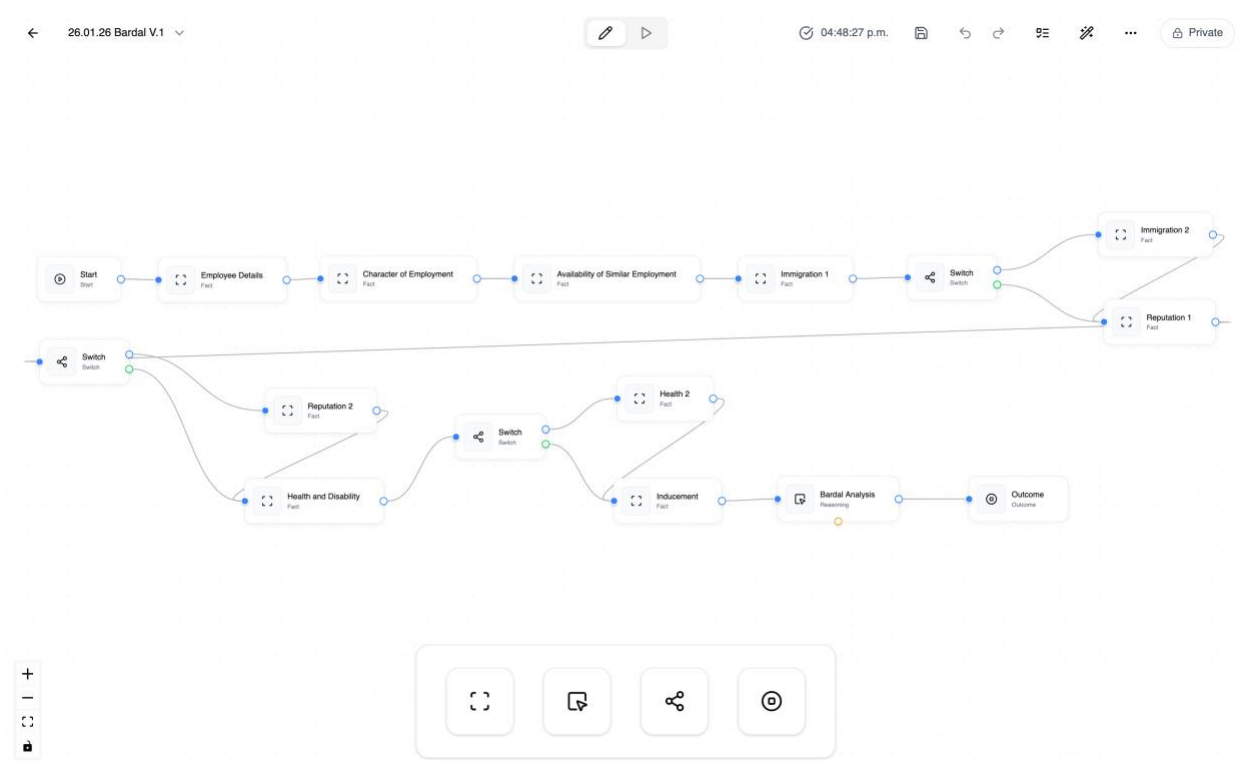
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Similar to our previous hackathons, the engineers could potentially collaborate with the lawyers and take their finished reasoning flows for building their own end-user tools. Any software developers interested in the project can team up with lawyers creating new reasoning flows or use our existing reasoning flows to integrate into a hypothetical workflow.

7. How the Benchmarkathon Works

1. Build — Encode Legal Reasoning

Participants create a **Reasoning Flow** describing how they analyze a recurring legal issue. This structured reasoning becomes a reusable guidance layer that AI agents must adhere and follow through before generating their final answer. Participants are encouraged to read over the ‘*OpenJustice User Guide*’ document for more instructional details on how to build reasoning flows.



2. Generate — Apply to Realistic Cases

Once your Reasoning Flow is complete, create at least five realistic test-case scenarios based on real-world use. Each scenario should include:

- **Input materials:** client statements, supporting documents, images (if applicable)

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- **Expected output:** the deliverable your Reasoning Flow should produce (e.g., demand letter, legal opinion)

The figure displays three sequential screenshots of the OpenJustice AI-automated benchmarking interface:

- 1. Create a New Evaluation Benchmark:** A form titled 'Create New Benchmark' with fields for 'Name' (e.g., 'Contract review accuracy'), 'Description' (e.g., 'What is this quality check meant to verify?'), and 'Dialog Flow Reference' (set to 'None'). Buttons for 'Cancel' and 'Create' are at the bottom.
- 2. Create Your Test Cases:** A form titled 'Edit Test Case' with a 'Name' field (e.g., 'Test Case X') and a large 'Input' text area containing a detailed legal scenario about an employee's dismissal and a request for a constructive dismissal claim. A 'Save Changes' button is at the bottom right.
- 3. Run Your Test Cases:** A form titled 'Run test case' showing the results of a test run. It includes a 'Which model generated these answers?' dropdown (set to 'GPT-3.5'), a 'Paste both AI answers for each test case' section, and a 'Start Comparison' button at the bottom right.

Figure 2: OpenJustice's AI-automated benchmarking feature

3. Evaluate — Compare Performance

This stage combines **model-based scoring (synthetic evaluation)** and **human expert review** to assess overall performance using the Legal AI Performance Evaluation Framework. Synthetic scoring ensures consistency and comparability, while human review validates legal soundness and strategic quality.

Performance is evaluated across the following criteria (which varies depending on transactional vs litigation tasks):

- **Citation Quality and Relevance**
- **Factual Accuracy and Hallucination Control**
- **Quality of Legal Reasoning**
- **Strategic Relevance of Analysis**
- **Conciseness and Non-Redundancy**
- **Document Structure and Formatting**
- **Completeness of Legal Coverage**
- **Consistency and Internal Coherence**
- **Appropriate Tone and Professional Register**

Together, these measures provide both quantitative benchmarking and expert qualitative validation, enabling reliable comparison across reasoning flows, prompts, and model iterations.

The diagram below illustrates how outputs from different AI systems are evaluated and compared under a unified performance framework.

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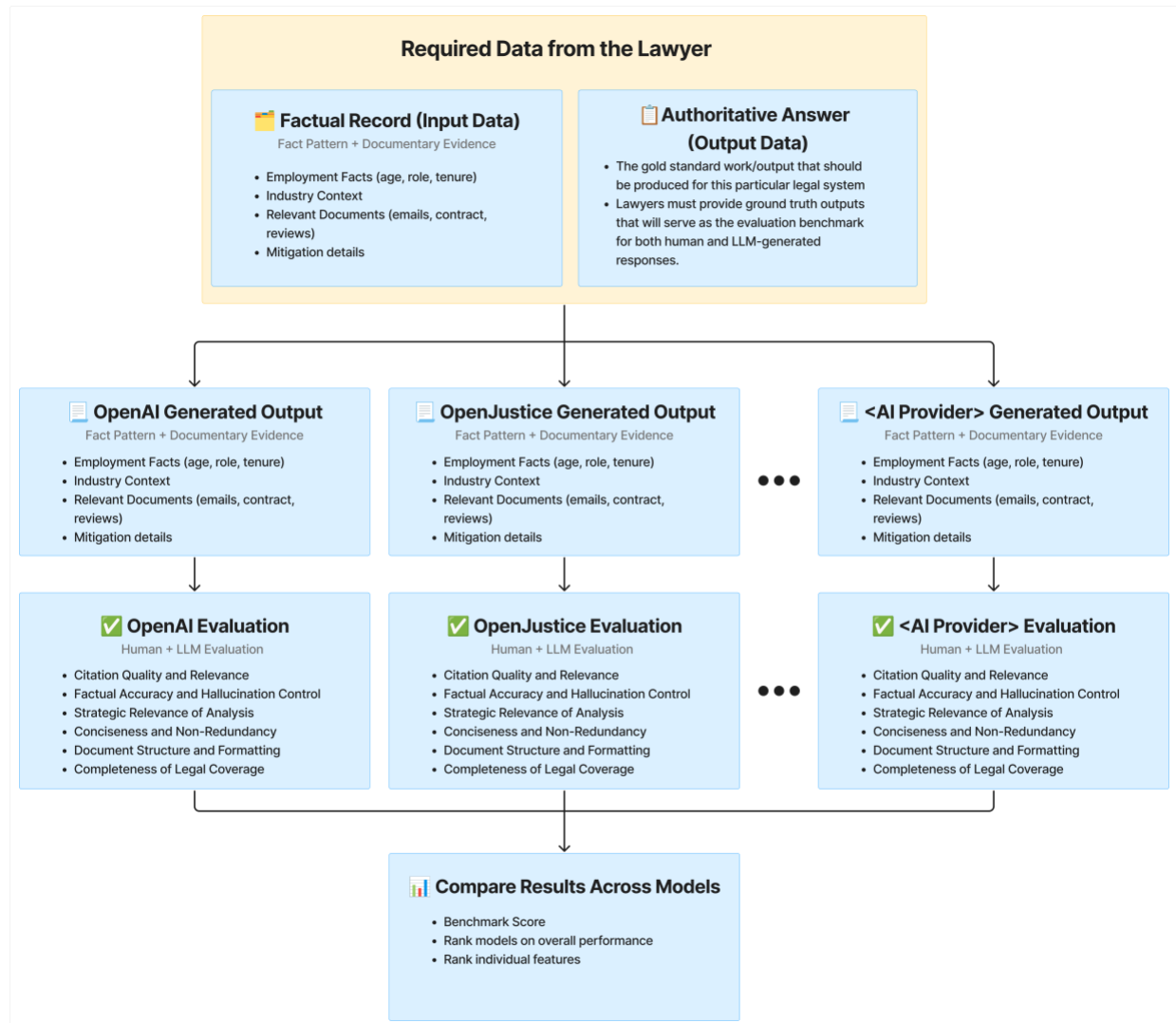


Figure 3: How Performance Is Measured. AI outputs are reviewed using a structured evaluation framework. Automated scoring ensures consistency, while human experts assess legal accuracy, reasoning quality, and professional judgment. Results are then compared across models

On OpenJustice, you can automate the evaluation of your Reasoning Flow responses and cross-compare outputs across AI models against a shared marking rubric. Once you're satisfied with your Reasoning Node's performance, use the AI-automated Benchmarking feature to measure how your model performs with and without your Reasoning Flow.

View our *OpenJustice User Guide* for more details.

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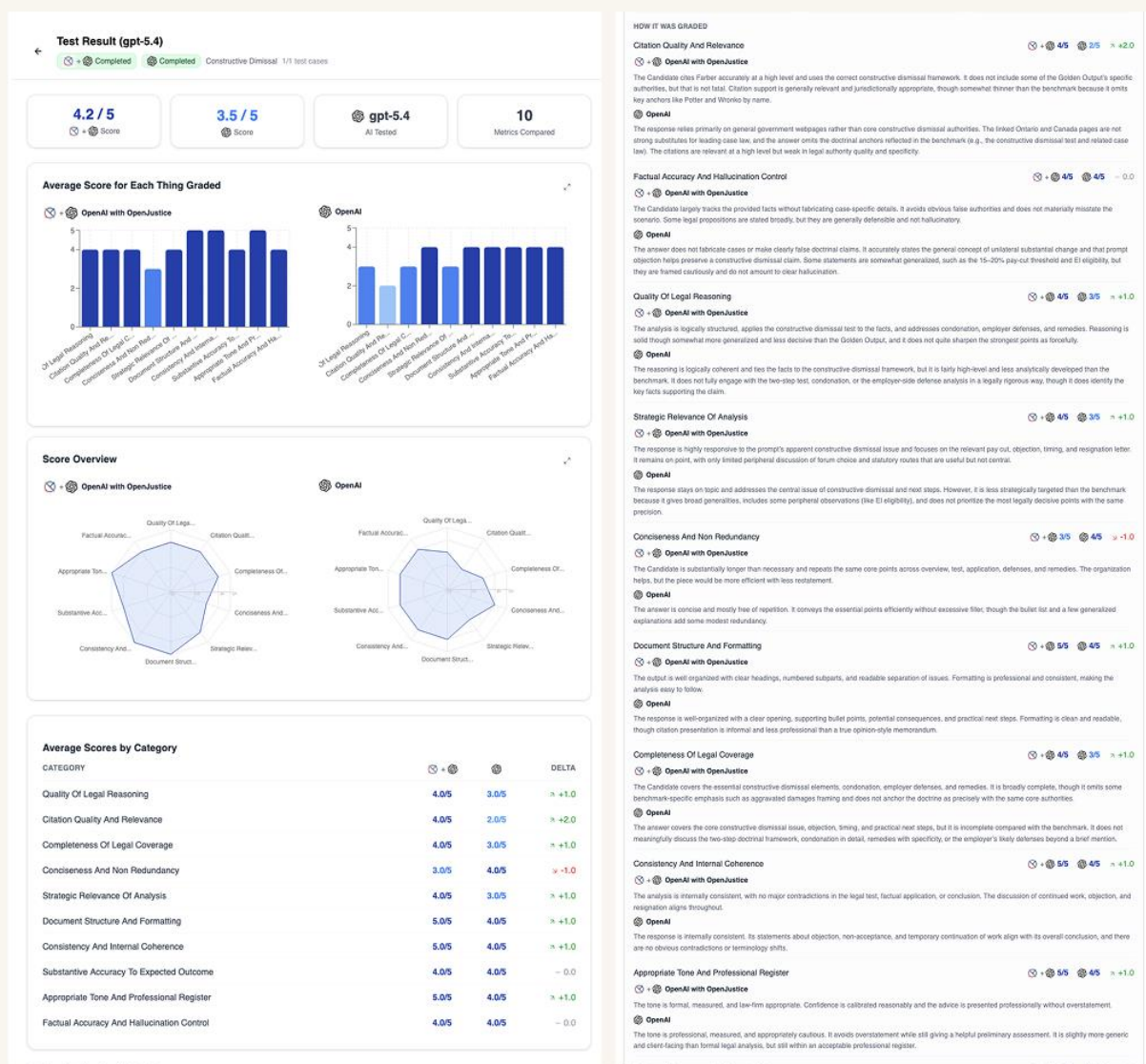


Figure 4: OpenJustice's AI-Automated Benchmarking Feature